DSA0502 – Query Processing for Data Science

CO5 AT3 – Visualization Design Activity

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7. Distribution Detective

Create a histogram of student marks or employee salaries. Identify the approximate shape, center, spread and possible skewness.

Expected Output: Histogram + observations

Aim

To create a histogram of student marks and identify the approximate shape, center, spread, and possible skewness of the distribution.

Dataset: student_marks.csv

```
%%writefile student_marks.csv
Student_ID,Marks
S01,45
S02,52
S03,55
S04,58
S05,60
S06,62
S07,64
S08,65
S09,65
S10,67
S11,68
S12,70
S13,72
S14,72
S15,74
S16,75
S17,75
S18,76
S19,76
S20,78
S21,78
S22,80
S23,82
S24,85
S25,88
S26,90
S27,91
S28,94
S29,96
S30,98
```

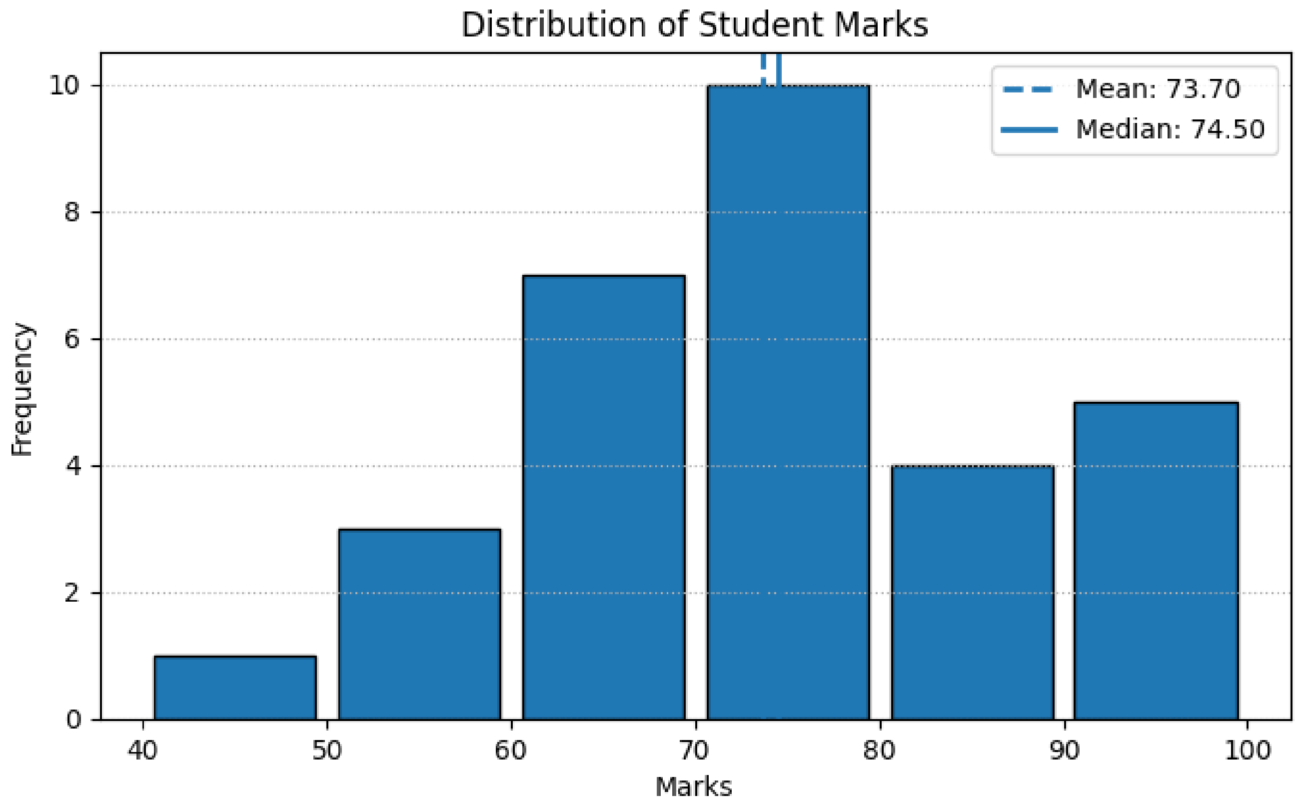
Google Colab Code

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import skew

df = pd.read_csv("student_marks.csv")

mean = df["Marks"].mean()
median = df["Marks"].median()
std = df["Marks"].std()
iqr = df["Marks"].quantile(0.75) - df["Marks"].quantile(0.25)
skewness = skew(df["Marks"])

print("Mean:", round(mean, 2))
print("Median:", round(median, 2))
print("Standard Deviation:", round(std, 2))
print("IQR:", round(iqr, 2))
print("Skewness:", round(skewness, 4))

plt.figure(figsize=(8, 4.5))
plt.hist(df["Marks"], bins=[40,50,60,70,80,90,100],
         edgecolor="black", rwidth=0.88)
plt.axvline(mean, linestyle="--", linewidth=2, label=f"Mean: {mean:.2f}")
plt.axvline(median, linestyle="-", linewidth=2, label=f"Median: {median:.2f}")
plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.legend()
plt.grid(axis="y", linestyle=":")
plt.show()
```

Observations

- Shape: Approximately unimodal and symmetric.
- Center: Mean and median are around the mid-70s.
- Spread: Marks range from 45 to 98; standard deviation and IQR show the amount of variation.
- Skewness: The value is close to zero, indicating very little skewness.

```
Mean: 73.7
Median: 74.5
Standard Deviation: 13.28
IQR: 16.5
Skewness: -0.0378
```



8. Counts and Frequencies

Create a categorical dataset such as department-wise employee count. Represent the frequencies using a bar chart.

Expected Output: Frequency chart

Aim

To create a categorical dataset representing department-wise employee counts and visualize the frequencies using a bar chart.

Dataset: employee_department.csv

```
%%writefile employee_department.csv
Department,Employee_Count
Engineering,120
Sales,85
Operations,65
Marketing,45
Finance,35
Human Resources,20
```

Google Colab Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("employee_department.csv")

df["Percentage"] = df["Employee_Count"] / df["Employee_Count"].sum() * 100
```

```
print("--- DEPARTMENT FREQUENCY TABLE ---")
print(df.to_string(index=False))

plt.figure(figsize=(8, 4.5))
bars = plt.bar(df["Department"], df["Employee_Count"], edgecolor="black")

for bar in bars:
    h = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, h + 2,
             str(int(h)), ha="center", fontweight="bold")

plt.title("Department-Wise Employee Count")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.xticks(rotation=20)
plt.grid(axis="y", linestyle=":")
plt.tight_layout()
plt.show()
```

Observations

- Highest frequency: Engineering has 120 employees.
- Lowest frequency: Human Resources has 20 employees.
- The bar chart clearly compares employee frequencies across departments.
- Engineering has the largest share of the total workforce.

```
--- DEPARTMENT FREQUENCY TABLE ---
     Department  Employee_Count  Percentage
    Engineering             120   32.432432
          Sales              85   22.972973
     Operations              65   17.567568
      Marketing              45   12.162162
        Finance              35    9.459459
Human Resources              20    5.405405
```